

Vitor Gama

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EDUCATION

West Virginia University

Ph.D. student, Chemical Engineering;

GPA: 3.75/4.00

Morgantown, WV, USA

Aug. 2021 – Dec. 2025 (Expected)

Federal University of Campina Grande

M.Sc., Chemical Engineering;

GPA: 9.14/10.00

Campina Grande, Paraiba, Brazil

Sept. 2018 – April 2021

Federal University of Campina Grande

B.Sc., Chemical Engineering;

GPA: 8.18/10.00

Campina Grande, Paraiba, Brazil

May 2013 – Aug. 2018

AWARDS & ACHIEVEMENTS

- **AIChE Environmental Division Graduate Paper Award (2024):** Awarded 2nd place with the paper titled “Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production”.
- **Jack and Marietta Mullenger Fellowship (2024)**
- **Statler College PhD Recruitment Fellowship (2021)**

RESEARCH PUBLICATIONS

Complete list available on my [Google Scholar](#).

- [1] **Vitor Gama et al.** “Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches”, **Industrial Engineering and Chemistry Research**. (under review).
- [2] **Vitor Gama et al.** “Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production”, **ACS Engineering Au**, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acsengineeringau.3c00069.
- [3] **Vitor V. Gama et al.** “Modeling and Process Operability Analysis of a Direct Air Capture System”, **IFAC-PapersOnLine**, 13th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2022, Vol. 55.7 (Jan. 2022), pp. 316-321. DOI: 10.1016/j.ifacol.2022.07.463. ISSN: 2405-8963.

CONFERENCE PRESENTATIONS

- [1] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Leveraging Process Operability Mapping to Support Experimental Membrane Direct Air Capture (m-DAC) Solutions*. 2024 AIChE Annual Meeting, San Diego, CA, October 29, 2024 (Oral Presentation)
- [2] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production*. Membranes: Materials and Processes – Gordon Research Conference, New London, NH, July 28 – August 2, 2024 (Poster Presentation)
- [3] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Evaluation of Hollow Fiber Membrane Modules for CO₂ Direct Air Capture and Utilization in Low-Purity CO₂ Processes*. 2023 AIChE Annual Meeting, Orlando, FL, November 7, 2023 (Oral Presentation)
- [4] **Vitor Gama**, San Dinh, Oishi Sanyal, Fernando V. Lima. *Feasibility Analysis of a Membrane System for Direct Air Capture of CO₂*. 2022 AIChE Annual Meeting, Phoenix, AZ, November 17, 2022 (Oral Presentation)

- [5] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Designing a Membrane-based Platform for CO₂ Direct Air Capture Systems Analysis*. Gordon Research Conference on Chemical Separations, Ventura, CA, October 2–7, 2022 (Poster Presentation)

RESEARCH EXPERIENCE

West Virginia University

Morgantown, WV, USA

Graduate Research Assistant (Ph.D.) [Advisors – Dr. Fernando V. Lima/Dr. Oishi Sanyal]

Aug. 2021 – Currently

- Process design & simulation
- Operability analysis studies to expedite process evaluation and deployment
- Application of operability and simulation frameworks to a gas separation membranes for Direct Air Capture (DAC) with Carbon Capture Utilization (CCU).

Federal University of Campina Grande

Campina Grande, Paraiba, Brazil

Graduate Research Assistant (M.Sc.) [Advisor - Dr. Antonio Tavernard]

Sept. 2018 – April 2021

- Chemical industry equipment cost estimation research based on literature models.
- Developed a Windows OS application for chemical industry equipment cost estimation based on literature models. (“CostApp: a Cost Estimation Tool Developed Using C# and WPF for the Chemical Engineering Field”)

Federal University of Campina Grande

Campina Grande, Paraiba, Brazil

Developer

Sept. 2019 – April 2021

- Technological Project for the Development of a Method for Synthesis of Control structures (PETROBRAS/CENPES/PAQTCPB/LENP/UFCG): An automated software capable of easily selecting the most promising self-optimizing control structures in industrial processes.
- Worked on coding the PID routines to control simulated processes uploaded to the BRPWC tool
- Worked on developing mock-ups for the process control user interface of the tool and code troubleshooting

SKILLS

Programming: Python, MATLAB, Markdown, C#, Javascript, LaTeX and exposure to Java

Technologies/Platforms: Git, GitHub, GitLab

Process simulation: Aspen Plus, Aspen Custom Modeler, HYSYS, AVEVA Process Simulation, PRO/II, ChemCad

Languages: English and Portuguese

RELEVANT COURSEWORK

Major coursework: Transport Phenomena, Advanced Chemical Engineering Thermodynamics, Chemical Reaction Engineering, Mathematical Methods in Chemical Engineering, Teaching Practicum

Minor coursework: Dynamic Simulations, Membrane Separations, Artificial Intelligence Techniques

PROFESSIONAL LEADERSHIP AND SERVICE

Brazilian Student Association (President)

Sept. 2022 – Sept. 2024

Statler College DEI Student Committee

May 2022 - May 2023

CODES Research Group Leader

Sept. 2022 - Present

CLIFTONSTRENGTHS

Arranger | Positivity | Input | Woo | Communication

COVER LETTER

Dear Prof. Stathatou,

My name is Vitor Gama, and I am currently a fourth-year Ph.D. candidate in the Chemical and Biomedical Engineering department at West Virginia University (WVU), working with Dr. Fernando V. Lima and Dr. Oishi Sanyal. My career in Chemical Engineering has been primarily focused on process systems engineering, particularly in process simulation and operability analysis. The postdoctoral position you announced on social media caught my attention, as I have recently been transitioning into the economic feasibility analysis of the membrane separation process that I have been studying throughout my Ph.D.

During my Ph.D. at WVU, I have been engaged in advancing the feasibility of gas separation membranes for direct air capture (m-DAC), particularly for low-purity CO₂ applications. My research bridges the gap between material properties and process-level performance by employing process operability and inverse design approaches. By systematically linking microscopic changes in facilitated transport membrane (FTM) parameters to macroscopic process outcomes, I have provided valuable insights into how membrane characteristics influence large-scale system efficiency. This work not only assesses the viability of m-DAC but also informs the design of optimized membranes tailored for real-world deployment. Below are two publications where I was the lead author:

- **Vitor Gama et al.**, “*Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches*”, Industrial Engineering and Chemistry Research (*under review*).
- **Vitor Gama et al.**, “*Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO₂ Production*”, ACS Engineering Au, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acseengineeringau.3c00069.

Beyond these publications, I am currently shifting my research focus toward the techno-economic analysis (TEA) of the m-DAC process. In parallel with this transition, I have been collaborating with Dr. Grigorios Pangakos and his team at Carnegie Mellon University (CMU) on coupling membrane and adsorption separation for DAC, using fluid dynamics and superstructure optimization strategies.

My body of work revolves around utilizing process simulation platforms, cost estimation techniques, open-source coding, and operability analysis to assess process feasibility. I firmly believe that the next steps of my research would align closely with your work on TEA and life cycle assessment (LCA), providing an opportunity for me to both contribute to your lab’s expertise and expand my knowledge in these essential fields of study.

In summary, working with your research group—given your strong background in process analysis, LCA, and TEA—would be an outstanding opportunity for me to share my knowledge and skills while gaining deeper insight into a new research area. I have attached my resume for your review, and I am happy to provide any additional details if needed. I look forward to hearing from you.

Best regards,

Vitor Gama

CONTACTS

- Dr. Oishi Sanyal (WVU) – oishi.sanyal@mail.wvu.edu
- Dr. Fernando V. Lima (WVU) – Fernando.Lima@mail.wvu.edu
- Dr. Heleno Bispo (UFCEG) – heleno.bispo@eq.ufcg.edu.br